SCHOOL OF AEROSPACE, MECHANICAL AND MECHATRONIC ENGINEERING

AERO2705: Space Engineering I

Satellite Orbits

Assignment 1

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1 Question 1: Mission Orbit Trade Study

1.1 Methodology

Three missions are evaluated, each treated as an unperturbed, Earth centred, circular orbit ($e = 0$): First, mission A is an Earth imaging satellite at 550 km altitude. Mission B, a GPS III class navigation satellite at 20 180 km altitude and Mission C is a meteorological satellite at 35 786 km altitude. We will be using the gravitation constant of $\mu = 398\,600\,\mathrm{km^3\,s^{-2}}$, radius of the earth $R_E = 6378\,\mathrm{km}$ and a sidereal day of 86 164 s in all questions.

For a satellite of negligible mass under the two-body point-mass assumption, meaning no drag, no third body or oblateness perturbation, gravity is equal to the centripetal acceleration for a circular orbit of radius $r = R_E + z$:

$$\frac{\mu}{r^2} = \frac{v^2}{r} \implies v = \sqrt{\frac{\mu}{r}}, \qquad T = \frac{2\pi r}{v} = 2\pi\sqrt{\frac{r^3}{\mu}}. \tag{1}$$

For in all the missions of this report, $r = R_E + z$ [km] is the orbital radius, μ [km^3/s^2] is Earth's gravitational parameter , v [km/s] is the orbital speed and T [s] is the orbital period. The orbital speed (v) sets the Δv that the launch vehicle must give to reach a given altitude. The period (T) shows how often each mission revisits a ground point. The two conserved quantities used throughout this report are the specific angular momentum $\mathbf{h} = \mathbf{r} \times \mathbf{v}$ [km^2/s] and specific orbital energy $\varepsilon = v^2/2 - \mu/r$ [km^2/s^2], follow from the unperturbed relative equation of motion $\ddot{\mathbf{r}} = -\mu\mathbf{r}/r^3$, in which $\mathbf{r}$ [km] are the position vectors and $\mathbf{a} = \ddot{\mathbf{r}}$ [km/s^2] the acceleration vectors of the satellite relative to centre of the earth. The angular momentum per unit mass ($\mathbf{h}$) points along the fixed orbit normal and its magnitude sets the orbit's size and shape (as $p = h^2/\mu$). ε is the mechanical energy per unit mass carried by the satellite, its sign distinguishes a bound orbit from an escape trajectory (will be used directly in the escape analysis of the last question of this report) and its magnitude is the quantity being traded off, mission by mission, throughout this report: dotting the equation of motion with $\mathbf{v}$ gives $\frac{d}{dt}\big(v^2/2 - \mu/r\big) = 0$, and crossing with $\mathbf{r}$ gives $\frac{d}{dt}(\mathbf{r} \times \mathbf{v}) = \mathbf{0}$, meaning that both are constant along the orbit. For a circular orbit $\mathbf{r} \perp \mathbf{v}$ everywhere, so $h = rv$ and substituting the velocity (1), $\varepsilon = v^2/2 - \mu/r = -\mu/2r = -\mu/2a$ (also using $a = r$, the semi major axis, that for a circle it is the same as the radius), this being consistent with the general result of $\varepsilon = -\mu/2a$ used for the eccentric orbits of Questions 2–4.

The values of angular momentum and orbital energy are confirmed by a second algebraically independent formula: h with $h = \sqrt{\mu p}$, and ε with $\varepsilon = v^2/2 - \mu/r$. As these routes are algebraically equivalent to the primary ones for $e = 0$ rather than independent physical models, the fact that lead to the same results means correct implementation, not the accuracy of the input altitudes or of μ, R_E themselves, as an error in either of the two would affect identically to both routes and cancel out of the residual.

Finally to put some numbers to have a quantitative feeling for part e, a simple horizon visibility geometry is added. An observer at ground sees a satellite once it clears the local horizon that for a spherical Earth occurs at an Earth centered half angle

$$\lambda = \arccos\left(\frac{R_E}{r}\right) \tag{2}$$

measured from the sub-satellite point, where λ [deg] is the Earth centred half angle and R_E, r are as defined before. The larger the λ means each satellite sees and is observed by a wider swath of Earth at once, this being why a higher orbit needs less satellites for the same coverage. Treating the visibility footprint as a spherical cap of half angle λ, its solid angle is $\Omega = 2\pi(1 - \cos\lambda)$, so a non overlapping packing argument gives a lower bound on the number of satellites needed to tile the whole sky at any instant:

$$N_{\min} = \left\lceil \frac{4\pi}{\Omega} \right\rceil = \left\lceil \frac{2}{1 - \cos\lambda} \right\rceil. \tag{3}$$

The 4π the solid angle of the whole sky as seen from the center of the earth, and Ω is the part of it covered by the satellite. Therefore N_{min} is the minimum satellite count for whole sky coverage. This is an idealization so it should not be taken into account for design purposes, just to have some quantities to compare.

1.2 Results and Discussion

Mission	Alt. [km]	r [km]	v [km/s]	T [h]	h [km^2/s]	ε [km^2/s^2]
A Earthimaging	550	6928	7.5852	1.5941	52 550	−28.767 32
B Navigation	20 180	26 558	3.8741	11.9647	102 888.4	−7.504 33
C Meteorological	35 786	42 164	3.0747	23.9343	129 640.2	−4.726 78

Table 1: Mission properties from Eq. (1) and $\varepsilon = -\mu/2r$.

We can see how radius and speed are inversaly proportional as seen in (1). This makes that Mission A that is the closest to Earth is the fastest ($v = 7.5852\,\text{km/s}$) with the shortest period ($T = 1.5941\,\text{h}$), and Mission C is the slowest ($v = 3.0747\,\text{km/s}$) with the longest period ($T = 23.9343\,\text{h}$). Angular momentum still increases with altitude because $h = rv \propto \sqrt{r}$ so the radius term dominates the diminishing speed. Every second route residual (h, ε for all the three missions) is at or below 1.5×10^{-11} against quantities of order of around 10^4 or 10^5, this being consistent with double precision round off rather than a modelling error.

Mission A completes 15.0142 orbits per sidereal day (quick and brief passes), Mission C gives a result of exactly 1.0000 this defines a geostationary orbit. Mission B stands out with 2.0004 orbits per sidereal day this being a semi synchronous 12h period, this is not a coincidence as 20 180 km is very close to the GPS semi synchronous altitude. Solving (1) for $T = 43\,082\,\text{s}$ (half a sidereal day) gives $r = 26\,561.7\,\text{km}$ that is only 3.7 km (0.014%) above Mission B radius, this small gap is why the value is 2.0004 rather than exactly 2.0000.

The energy ladder is:

$$\Delta\varepsilon_{A\to B} = 21.263\,\text{km}^2/\text{s}^2, \quad \Delta\varepsilon_{B\to C} = 2.778\,\text{km}^2/\text{s}^2, \quad \Delta\varepsilon_{A\to C} = 24.041\,\text{km}^2/\text{s}^2 = 83.57\%\text{ of } |\varepsilon_A|. \tag{4}$$

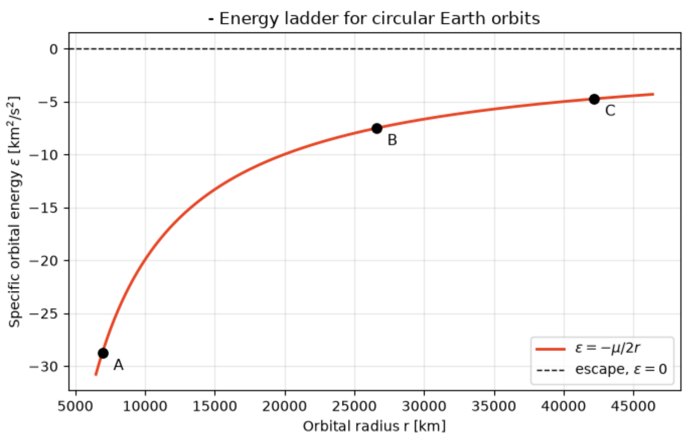

Figure 1: Specific orbital energy ε vs radius r.

The A to B altitude gain (19 630 km) is actually a bit larger than the B to C gain (15 606 km), but the B to C energy increase is less than one eighth of the A to B increase because $\varepsilon(r) = -\mu/2r$ is not linear, its slope $d\varepsilon/dr = \mu/2r^2$ falls with the square of the radius. The average slope over each segment is 1.0832×10^{-3} and $1.7798 \times 10^{-4}\,\text{km/s}^2$ respectively, a factor of 6.09, confirming the quadratic decrease in energy. This meaning that near earth a small climb requires a really large increment of energy needed to escape, but far away, climbing barely raises ε per km, which is why extending Mission B's orbit out to GEO costs so little compared with the climb from LEO to MEO.

No mission dominates in all metrics. Mission A has the least specific energy ($\varepsilon_A = -28.767\,\text{km}^2/\text{s}^2$), so it is cheapest to orbit and gives the lowest latency as its closer, but its 1.5941 h period means any ground point is visible for a short period of time only. Its horizon half angle from (2) is only $\lambda_A = 22.98°$, giving $N_{\min} = 26$ from (3) therefore a large constellation of satellites is required for global coverage. Mission C has the most energy ($24.041\,\text{km}^2/\text{s}^2$ more than Mission A) but also has the longest signal path (42 164 km), but its exact 1 orbit per day period gives continuous single satellite dwell over one hemisphere, that it is ideal for continuous meteorological monitoring, the $N_{\min}$ bound does not apply here in the same way as here a single geostationary satellite already provides fixed, continuous coverage. Mission B is considered the best for the amount of energy invested as it achieves $\lambda_B = 76.10°$ and $N_{\min} = 3$, one order of magnitude less in required satellites when compared to Mission A, also a repeating 12h ground track is perfect for navigation and all this for only 88% of the energy required to reach GEO (21.263 instead of the 24.041 km^2/s^2 climb from A to C). The λ and $N_{\min}$ shows that climbing from A to B provides a massive coverage expansion while energy is still relatively cheap per km, but continuing from B to C yields a negligible coverage gain (from 76.10° to 81.30° with the same $N_{\min} = 3$) for an additional 12% of energy expended.

2 Question 2: Molniya Orbit Design and Numerical Propagation

2.1 Methodology

The orbit is stated as semi synchronous, meaning two orbits per sidereal day, with a perigee altitude of 500 km. Therefore the period required is half a sidereal day, $T = 86\,164\,\text{s}/2 = 43\,082\,\text{s}$, from which (1) gives the semi-major axis $a = (\mu T^2/4\pi^2)^{1/3}$. With $r_p = R_E + 500 = 6878\,\text{km}$ and the rest of geometric elements being:

$$
\begin{aligned}
a &= \left(\frac{\mu T^2}{4\pi^2}\right)^{1/3} = \left(\frac{398600 \times 43082^2}{4\pi^2}\right)^{1/3} = 26\,561.7\,\text{km},\\
r_a &= 2a - r_p = 2(26561.7) - 6878 = 46\,245.5\,\text{km},\\
e &= \frac{r_a - r_p}{r_a + r_p} = \frac{46245.5 - 6878}{46245.5 + 6878} = 0.7411,\\
p &= a(1-e^2) = 26561.7\left(1 - 0.7411^2\right) = 11\,974.98\,\text{km},\\
h &= \sqrt{\mu p} = \sqrt{398600 \times 11975} = 69\,088.56\,\text{km}^2/\text{s}.
\end{aligned} \tag{5}
$$

Here r_a [km] is the apogee radius, e is the eccentricity (that is 0 for a circle and from 0 to 1 for an ellipse) and p [km] is the semi-latus rectum that is the radius at 90° true anomaly. h [km^2/s] is the specific angular momentum, the same as in Question 1. A large e is what this mission needs as this will let the satellite to stay some time close to the apogee, that will be quantified later with the Kepler's Second Law, while still meeting the 500 km altitude constraint at perigee. Because velocity is purely tangential at both apsides, $v_p = h/r_p$ and $v_a = h/r_a$ are the perigee and apogee speeds [km/s], v_p fixes the initial condition for the propagation below, and the difference between v_p and v_a is whay we have a long apogee dwell. For a conic section, evaluating the constant specific energy at perigee $\varepsilon = v^2/2 - \mu/r$ with $v_p = h/r_p$, $h^2 = \mu a(1-e^2) = \mu r_p(1+e)$ and $r_p = a(1-e)$ results in:

$$
\varepsilon = \frac{\mu(1+e)}{2r_p} - \frac{\mu}{r_p} = \frac{\mu(e-1)}{2r_p} = -\frac{\mu(1-e)}{2a(1-e)} = -\frac{\mu}{2a}, \tag{6}
$$

generalising the circular orbit result of Question 1 to any eccentricity as ε depends only on a now and it does not depend on e, so this is the background of the direct comparison with Mission B for Section 2.2.

For the next part, the orbit is propagated from the state at perigee $\mathbf{y}_0 = [r_p, 0, 0, 0, v_p, 0]^\mathsf{T}$ where we can see how the motion is confined to the 2D plane since the two body force has no out of plane component, using a fourth order Runge Kutta (RK4) integrator applied to this first order system $\dot{\mathbf{y}} = f(\mathbf{y}) = [\mathbf{v};\, -\mu\mathbf{r}/r^3]$, which does not depend on time. Here $\mathbf{y}$ is the state vector, that contains position and velocity all in one vector, therefore it fully describes the satellite at an instant, and f is the dynamics function that returns its time derivative, so that we can propagate $\mathbf{y}$ forward in time so that this produces the future trajectory. Each step of the RK4 is defined as:

$$\mathbf{k}_1 = f(\mathbf{y}_i), \quad \mathbf{k}_2 = f(\mathbf{y}_i + \tfrac{\Delta t}{2}\mathbf{k}_1), \quad \mathbf{k}_3 = f(\mathbf{y}_i + \tfrac{\Delta t}{2}\mathbf{k}_2), \quad \mathbf{k}_4 = f(\mathbf{y}_i + \Delta t\,\mathbf{k}_3),$$
$$\mathbf{y}_{i+1} = \mathbf{y}_i + \frac{\Delta t}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4). \tag{7}$$

The $\mathbf{k}$ here are four trial estimates of the state's rate of change across the step, at the start, twice at the midpoint using the previous trial, and at the projected endpoint, and Δt [s] is the fixed timestep. After all 4 $\mathbf{k}$ calculated, their weighted average is what makes $\mathbf{y}_i$ to $\mathbf{y}_{i+1}$ with local error of $O(\Delta t^5)$, that is much smaller that the one if you use single slope Euler estimate $\mathbf{k}_1$ alone. This is why we preffer to use RK4 instead of a lower order method as this allows us to keep Δt practically large without lowering the accuracy.

Because $T/\Delta t$ is not an integer for the chosen time step, a fixed step loop would either stop short of T or overshoot it so the implementation instead takes $n = \lceil T/\Delta t \rceil$ steps and shortens the final step to $\min(\Delta t,\, T - t_i)$, making the last step to land exactly at $t = T$. This matters directly for the return error check in the last part of the question. Without this, the check would be comparing positions at two different times rather than testing the integrator itself as they wouldn't be finishing at exactly the same time.

We will be checking the timesteps by convergence, Table **??** are the results of propagating the same orbit at this different time steps $\Delta t \in \{240, 120, 60, 30, 10, 5, 1\}$ s and checks the maximum relative drift of h and ε and the return error in km after one period. From the converged trajectory the magnitude of $\mathbf{r} \times \mathbf{v}$ and of $v^2/2 - \mu/r$ are recomputed at each sample and are compared with their initial values, the swept area between consecutive samples:

$$\Delta A_i \approx \tfrac{1}{2}|\mathbf{r}_i \times \mathbf{r}_{i+1}|,$$

where $\mathbf{r}_i$ [km] is the propagated position at sample i and ΔA_i [km^2] is the area swept by consecutive samples that divided by the time step gives the areal rate. Altitude is then extracted at every sample to find the fraction of the period spent above 20 000 km and the distance between the final and initial propagated positions gives the return error.

2.2 Results and Discussion

Quantity	Value
Period T	43 082 s
Semi-major axis a	26 561.7 km
Apogee radius r_a (altitude)	46245.5 (39867.5) km
Eccentricity e	0.7411
Semi-latus rectum p	11 974.98 km
Specific angular momentum h	69 088.6 km^2/s
Perigee / apogee speed v_p, v_a	10.0449 / 1.4940 km/s
Specific orbital energy ε	-7.5033 km^2/s^2

Table 2: Design and propagation results for the Molniya orbit.

This ε almost matches Mission B from Question 1 ($-7.504\,33$ km^2/s^2). This is because both orbits are semi-synchronous so they share the same period and therefore, by (6), the same semi major axis and the same energy, energy depends only on a, not on e, so that the same energy buys a near circular MEO orbit or a highly eccentric Molniya orbit. Then, period and energy are locked together, but shape is a free design choice. The small residual difference (0.001 05 km^2/s^2, 0.014%)

is the same 3.7 km gap that in Question 1 between Mission B real world altitude and the exact semi-synchronous radius, this orbit was built directly from the period requirement so it sits exactly on the semi-synchronous value without this gap.

Δt [s]	steps	max. drift h	max. drift ε	return err. [km]
240	180	2.907×10^{-5}	5.186×10^{-4}	336.274
120	360	1.000×10^{-6}	2.381×10^{-5}	15.443
60	719	3.131×10^{-8}	1.224×10^{-6}	0.794
30	1437	9.792×10^{-10}	6.842×10^{-8}	0.044 37
10	4309	4.043×10^{-12}	7.790×10^{-10}	0.000 51

Table 3: RK4 step-size convergence over one propagated period.

Halving the step from 60 s to 30 s improves the h drift by 32 times and the return error by 18 times. Lowering the timestep more to $\Delta t = 10$ s lowers another order of magnitude but triples the step count. $\Delta t = 30$ s (1437 steps) is used as the converged working step as it is a good trade between accuracy and runtime. From the table we can also see that the integrator works fine as a fourth order method since RK4 global error scales as $O(\Delta t^4)$, halving Δt should shrink the return error by a factor near $2^4 = 16$. The observed ratios are $336.274/15.443 = 21.8$ (240 to 120 s), $15.443/0.794 = 19.4$ (120 to 60 s) and $0.794/0.04437 = 17.9$ (60 to 30 s) therefore converging toward 16 as Δt shrinks and the step size becomes small enough for the leading order error term to dominate, which is the expected behaviour of a fourth order method. At the chosen step the maximum relative drift is 9.792×10^{-10} in h and 6.842×10^{-8} in ε, both essentially conserved, and the mean areal rate from successive samples, 34 543.914 km^2/s, matches almost exactly the theoretical $h/2 = 34\,544.278$ km^2/s, directly verifying Kepler's Second Law, equal area sweeps in equal times. This conservation is a consequence of the high order of accuracy of RK4 over the single period integrated here, not a structural guarantee, nothing forces h or ε to be conserved exactly at any of step size and over many more periods a slow secular drift would eventually appear. The close to machine precision seen here should be taken as an evidence of an accurate integration over this orbit not as an evidence that this particular RK4 implementation conserves energy for any long propagations.

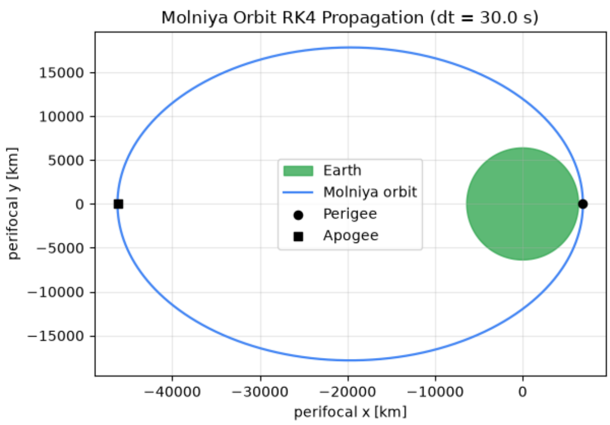

Figure 2: Propagated Molniya orbit.

The perigee speed is 10.0449 km/s compared to an apogee speed of only 1.4940 km/s, a factor of 6.72 slower, this giving direct evidence of Kepler's Second Law since $h = rv \sin \gamma$ is fixed and $v_a/v_p = r_p/r_a$ at the tangential apsides. The satellite spends 73.88 % of the period above 20 000 km altitude.

This long, slow apogee dwell, positioned over the northern hemisphere by the inclination and orientation is what makes the orbit good for high latitude communications as a ground station sees one satellite at high elevation for most of each half period. Under the idealised assumption of an instantaneous handover between satellites with no minimum elevation margin, continuous coverage needs $\lceil 1/0.7388 \rceil = 2$ satellites in suitably phased orbits. This is a lower bound, in real Molniya constellations historically flew more satellites to allow handover margins.

The propagated position after one period differs from the initial position by 0.044 37 km. This is due to truncation error, not because of the endpoint handling described above, since the final sample is defined to land exactly at $t = T$. The return error check is what tests timing and it confirms the satellite arrives back at its starting point at $t = T$ to within 44 m, showing that the integrator advances time correctly and not just that it stays on the correct ellipse.

3 Question 3: Satellite Orbit Determination

3.1 Methodology

From this two perifocal observations that are given, first one with a perigee state $\mathbf{r}_1 = [7000, 0, 0]$ km, $\mathbf{v}_1 = [0, 8.0, 0]$ km/s, and a second position $\mathbf{r}_2 = [-6097.5, 6097.5, 0]$ km at an unknown true anomaly that is the angle in the direction of motion measured from perigee to where the satellite sits on its orbit. From $\mathbf{r}_1, \mathbf{v}_1$ the specific angular momentum follows as $\mathbf{h} = \mathbf{r}_1 \times \mathbf{v}_1$, exactly as defined in Question 1.

The eccentricity vector is defined as

$$\mathbf{e} \equiv \frac{\mathbf{v} \times \mathbf{h}}{\mu} - \frac{\mathbf{r}}{r}, \tag{8}$$

and is, together with $\mathbf{h}$ and ε, a conserved quantity of the two-body equation of motion $\ddot{\mathbf{r}} = -\mu \mathbf{r}/r^3$ (Question 1, Eq. 3). By definition, $\mathbf{e}$ points from the focus toward perigee, with magnitude equal to the orbit's eccentricity, so a single conserved vector fixes both the orbit's orientation and its shape. At perigee $\mathbf{r}_1$ is parallel to $\mathbf{e}$ by definition, so (8) reduces to the scalar $e = (h^2/\mu - r_1)/r_1$. Then, we can get the semi-major axis from $\varepsilon = v_1^2/2 - \mu/r_1 = -\mu/2a$, and the period from (1). The orbit is classified in two different ways: geometrically from e, being an ellipse if $0 < e < 1$, and energetically depending on the sign of ε, being bound if negative.

The true anomaly at $\mathbf{r}_2$ is found by two ways. First directly from the direction of $\mathbf{r}_2$ in the perifocal frame ($\theta = \operatorname{atan2}(r_{2y}, r_{2x})$, valid since $\mathbf{r}_1$ defines $\theta = 0$ at perigee), and secondly by solving the orbit equation

$$r = \frac{p}{1 + e\cos\theta}$$

for θ at $r = |\mathbf{r}_2|$, both as stated in the question. This is the polar equation of a conic section with the focus at the origin, where r [km] is the orbital radius at true anomaly θ, and $p = h^2/\mu$ [km] is the semi-latus rectum already introduced in Question 2, the same equation used there is used here but in the opposite direction, now to solve for θ from a known r. After this, we will check the result of θ from the direction of $\mathbf{r}_2$ substituting it back into the orbit equation to predict r, and then compare it with $|\mathbf{r}_2|$ directly.

Propagating a true anomaly change $\Delta\theta = 60°$ from perigee uses the Lagrange coefficients, each computed from its own expression with h, μ, r_1 and the target radius $r_3 = p/(1 + e\cos\Delta\theta)$:

$$\begin{gathered} f = 1 - \frac{\mu r_3}{h^2}(1 - \cos\Delta\theta), \qquad g = \frac{r_3 r_1}{h}\sin\Delta\theta, \\ \dot{f} = \frac{\mu}{h}\frac{1-\cos\Delta\theta}{\sin\Delta\theta}\left[\frac{\mu}{h^2}(1-\cos\Delta\theta) - \frac{1}{r_1} - \frac{1}{r_3}\right], \qquad \dot{g} = 1 - \frac{\mu r_1}{h^2}(1 - \cos\Delta\theta), \end{gathered} \tag{9}$$

$f, \dot{g}$ [dimensionless] and g [s], $\dot{f}$ [s^{-1}] are scalar coefficients that let the position and velocity at the new true anomaly be written directly as linear combinations of the known state at $\mathbf{r}_1, \mathbf{v}_1$, without solving Kepler's time of flight equation, and this is what makes propagation by a specified angle (rather than a specified time) a closed form calculation instead of an iterative one. This gives the propagated state $\mathbf{r}_3 = f\mathbf{r}_1 + g\mathbf{v}_1$, $\mathbf{v}_3 = \dot{f}\,\mathbf{r}_1 + \dot{g}\,\mathbf{v}_1$.

After this, it is validated with three independent ways: firstly comparing $|\mathbf{r}_3|$ against the orbit equation evaluated at $\theta = 60°$, secondly $|\mathbf{v}_3|$ against the speed from the general vis-viva equation, $v = \sqrt{\mu(2/r - 1/a)}$, that is the same energy relation as the one used in (6), rearranged for speed at any radius r on the orbit rather than only at the apsides as used in Question 2, now evaluated at r_3, and lastly the identity $f\dot{g} - \dot{f}g = 1$ that holds for any correctly evaluated set of Lagrange coefficients independently of the actual parameters of the orbit.

3.2 Results and Discussion

Quantity	Value	Units
h	56000.00	km^2/s
e	0.123934	–
a	7990.2635	km
T	7108.09 (1.9745)	s (h)
ε	-24.942857	km^2/s^2

Table 4: Determined orbital elements and propagated state.

With $0 < e < 1$ and $\varepsilon < 0$ the orbit is classified an ellipse by both independent routes, as required. The true anomaly at Position 2 is $\theta = 135.0°$ from the direction of $\mathbf{r}_2$, compared to the 134.9958° from the orbit equation showing a difference of 0.0042°, consistent with the position coordinates given as they where supplied with just one decimal place instead than as an exact conic point. Substituting the calculated θ back into the orbit equation we get $r = 8623.2278$ km compared to the given $|\mathbf{r}_2| = 8623.1672$ km, giving a residual of 60.6 m, the same rounding, propagated through a different equation, so the two checks are not independent evidence of the same error but two different views of it.

Propagating $\Delta\theta = 60°$ from perigee gives $f = 0.5291755$, $g = 801.9890$ s, $\dot{f} = -8.8060644 \times 10^{-4}$ 1/s, $\dot{g} = 0.5551339$, therefore:

$$\mathbf{r}_3 = [3704.2287,\ 6415.9123,\ 0]\ \text{km}, \qquad \mathbf{v}_3 = [-6.16425,\ 4.44107,\ 0]\ \text{km/s}.$$

We can see how all of the three validation checks are satisfied with precision, firstly $|\mathbf{r}_3| = 7408.4574$ km matches the orbit equation at $\theta = 60°$ exactly to 4 decimal places, secondly $|\mathbf{v}_3| = 7.597\,436$ km/s matches the vis-viva speed to 6 decimal places and $f\dot{g} - \dot{f}g = 1$ exactly. Each check shows and probes a different failure mode, the first one with the radius check validates only the propagated position's magnitude, not its direction or the velocity at all, the second one, the vis-viva check validates only the propagated speed and not its direction and finally, the identity validates that the four Lagrange coefficients are consistent,it would equal 1 even if h, e or $\Delta\theta$ had been entered incorrectly, since it is a property of $f, g, \dot{f}, \dot{g}$, not a check against the physical orbit. Only the three checks together constrain both the magnitude and the physical validity of the propagated state.

4 Question 4: Historical Mission Analysis: Chandrayaan-3

4.1 Introduction

This mission was launched by ISRO on 14 July 2023 and reached the Moon by raising its Earth orbit through repeated perigee burns before a single trans-lunar injection, this differenciates it to the Apollo missions with their single and powerful translunar injection just after launch. Chandrayaan-3 took 40 days to go from Earth to landing, one order of magnitude larger than Apollo's mission that took about 3 days from launch to the lunar orbit. This section talks about the manoeuver sequence with sourced dates and altitudes, computes the two-body orbital elements of the parking and transfer orbits, also tracks the specific energy ladder the spacecraft climbed across five Earth bounded manoeuvres, and compares a two-body transfer time estimate with the mission's actual timeline to see how much of the difference is explained only by orbital mechanics.

4.2 Methodology

The mission parameters were obtained from ISRO status updates that can be looked up at the Press Information Bureau of the Government of India and by the eoPortal satellite missions directory [1,2], an initial 170 km × 36 500 km parking orbit at launch, followed by Earth bound manoeuvres (EBM) on 15, 20 and 25 July raising the orbit to 173×41 762 km, 233×71 351 km and 236×127 603 km respectively (expressed as perigee altitude × apogee altitude), then a trans-lunar injection the 31 July/1 August 2023 into a 288×369 328 km transfer orbit, a lunar orbit insertion (LOI) on 5 August 2023 and landing on 23 August 2023.

For each documented orbit, altitudes are are used to elements exactly as in Questions 1 to 3: $a = \frac{1}{2}(r_p + r_a)$ [km] the semi-major axis from the two apsis radii r_p, r_a, $e = (r_a - r_p)/(r_a + r_p)$ the eccentricity, $\varepsilon = -\mu/2a$ [km^2/s^2] specific orbital energy from Eq. 6, $T = 2\pi\sqrt{a^3/\mu}$ [s] period from Eq. 1 and, for the parking and transfer orbits, the perigee speed $v_p = \sqrt{\mu(2/r_p - 1/a)}$ [km/s] from the general vis-viva equation introduced in Question 3. Reusing this set of formulas across every documented orbit makes the energy values in Table 5 directly comparable to each other and to the missions of Question 1. The energy of every documented orbit is plotted against the manoeuvre sequence, and the total climb from parking orbit to transfer orbit is compared with the energy still needed to reach escape ($\varepsilon = 0$) from the parking orbit. Escape speed at the transfer orbit's perigee altitude,

$$v_{\text{esc}} = \sqrt{\frac{2\mu}{r_p}},$$

is the minimum speed [km/s] at radius r_p for $\varepsilon \geq 0$, i.e. for the spacecraft to leave Earth's gravity well entirely instead of fall back, so we can set $\varepsilon = 0$ in $\varepsilon = v^2/2 - \mu/r$ and solve for v. It is compared with the actual perigee speed there, and the characteristic energy

$$C_3 = 2\varepsilon$$

of the transfer orbit is reported, C_3 [km^2/s^2] is the same specific energy already used throughout this report but simply doubled and renamed to match the convention launch vehicle performance charts use to state how much energy a payload can be given beyond a circular parking orbit, its sign here that it is negative, since the orbit is bound, is the proof of whether the spacecraft was actually sent toward the Moon on a bound transfer or had it been positive on an escaping hyperbolic path. Finally, the transfer orbit's half period is compared with the actual time from TLI to LOI, using the timestamps above (TLI ≈ 00:15 IST, 1 Aug. = 18:45 UTC, 31 Jul.; LOI ≈ 19:00 IST, 5 Aug. = 13:30 UTC, 5 Aug., both as announced by ISRO [3]).

4.3 Results and Discussion

Orbit (date)	a [km]	e	ε [km^2/s^2]	T [days]
Parking (14 Jul)	24 713.0	0.735 04	−8.064 58	0.4475
EBM-1 (15 Jul)	27 345.5	0.760 44	−7.288 22	0.5209
EBM-4 (20 Jul)	42 170.0	0.843 23	−4.726 11	0.9975
EBM-5 (25 Jul)	70 297.5	0.905 91	−2.835 09	2.1469
TLI transfer (31 Jul/1 Aug)	191 186.0	0.965 13	−1.042 44	9.6290

Table 5: Chandrayaan-3 Earth departure orbits.

The parking orbit has perigee speed v_p = 10.277 06 km/s, then the transfer orbit of v_p = 10.840 07 km/s at its 288 km perigee altitude.

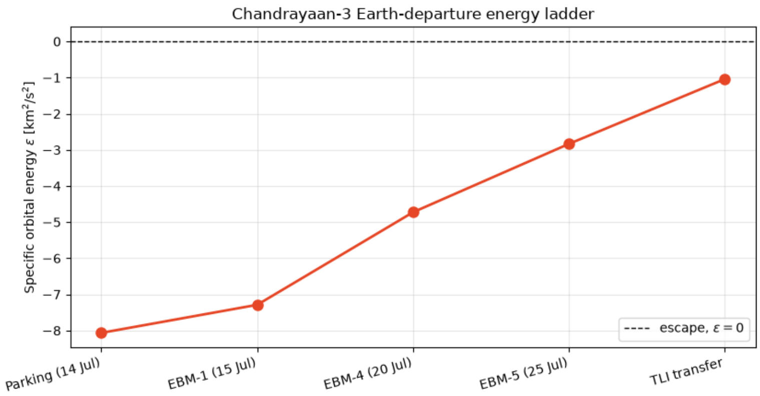

Figure 3: Specific orbital energy at each documented manoeuvre.

The spacecraft gained 7.0220 $\mathrm{km^2/s^2}$ of specific energy from parking orbit to transfer orbit, against 8.0646 $\mathrm{km^2/s^2}$ that would have been needed to reach escape from the parking orbit 87.07 % of the way there. Fig. 3 shows that the first two EBMs (from 15th to 20th July) together contribute less than half the total energy gain, while the last two steps (EBM-5 and TLI) contribute the rest, because $\varepsilon = -\mu/2a$ flattens as a grows, the same reasoning as in the Question 1 energy ladder, each further perigee burn buys progressively more Δa per unit of $\Delta\varepsilon$, so late stage burns look dramatic in apogee altitude but they are very efficient in energy terms.

At the transfer orbit at 288 km altitude perigee ($r_p = 6666$ km), escape speed is $v_{\text{esc}} = \sqrt{2\mu/r_p} =$ 10.9358 km/s, against the actual perigee speed of 10.840 07 km/s – 0.095 74 km/s short of escape. The characteristic energy is $C_3 = 2\varepsilon = -2.084\,88\,\mathrm{km^2/s^2}$ therefore matching $v_p^2 - 2\mu/r_p$ to 5 significant figures. Its negative sign confirms the transfer orbit is bound (elliptical), as if C_3 had been positive, the perigee burn would have exceeded local escape speed and the spacecraft would have departed Earth on a hyperbolic trajectory with hyperbolic excess speed $v_\infty = \sqrt{C_3}$ therefore never to returning to a closed Earth orbit, the opposite of the staged, bound approach that was actually flown.

The half-period of the transfer orbit is $T/2 = 4.8145$ days and the actual time from TLI to LOI, using the timestamps in Section 4.2, is 4.7813 days, agreement to 0.0333 days (0.69 %). This agreement shows the coast from TLI to the proximity of the Moon is well described by an unperturbed two-body Earth ellipse for almost its full duration, with the Moon's gravity only altering the trajectory significantly at the final approach, inside the lunar sphere of influence (around 66 000 km from the Moon), a small fraction of the 369 328 km transfer apogee. The two-body model therefore gives out this late and localised perturbation and not a gradual drift accumulated over the whole transfer. If, instead, the discrepancy had been on the order of tens of percent, that would point to a transfer geometry poorly approximated by a simple ellipse rather than to a short late stage effect.

The ISRO's staged, multi perigee burn strategy contrasts with Apollo's single, powerful translunar injection within hours of launch. LVM3, Chandrayaan-3's launch vehicle, and the propulsion module's engine are both much less powerful than the Saturn V/S-IVB stage that sent Apollo directly into a translunar trajectory, this delivering the same total Δv in one burn was not available, so it was instead delivered in instalments using the spacecraft's smaller engine and propellant budget when compared to Apollo's. Firing specifically at perigee, where speed is highest, is the efficient choice by the Oberth effect that says that a fixed Δv applied at high speed produces a bigger change in specific energy ($\Delta\varepsilon \approx v\,\Delta v$ for a tangential burn) than the same Δv applied near apogee, where v is much lower, which is also why each of the three EBMs in Table 5 was timed at perigee rather than spread continuously around the orbit.

Date (2023)	Event	Resulting orbit
14 Jul	Launch (LVM3), parking orbit injection	170×36500 km (Earth)
15 Jul	EBM-1 (Earth-bound perigee firing)	173×41762 km
20 Jul	EBM-4	233×71351 km
25 Jul	EBM-5 (final Earth-bound manoeuvre)	236×127603 km
31 Jul/1 Aug	Trans-lunar injection (TLI)	288×369328 km
5 Aug	Lunar orbit insertion (LOI)	164×18074 km (lunar)
23 Aug	Landing (Vikram lander, south polar region)	–

Table 6: Chandrayaan-3 mission timeline [1–3].

4.4 Conclusion

Treating each documented altitude pair as a two-body conic reproduces the mission's own account of its trajectory to the precision of the publicly reported altitudes, so we can observe that the parking to transfer energy climb accounts for 87.07 % of the specific energy needed to reach escape, the transfer orbit's perigee speed falls 0.0957 km/s short of local escape speed meaning that it has a bound, not an hyperbolic departure, and the unperturbed half-period of that transfer orbit predicts the TLI to LOI coast time to within 0.69 %. This last result is the most informative because the two-body model contains no lunar gravity at all, its close agreement with the actual flight time shows that the Moon's influence is just important at the short final phase near the lunar sphere of influence rather than being spread across the transfer, which is exactly the regime in which a patched conic, two-body approximation is expected to work well. The staged, multi perigee burn strategy is explainedby the practical mismatch between LVM3's injection capability and the total energy needed for translunar transfer, which the spacecraft's own small propulsion module made up in five Oberth efficient instalments at perigee rather than in one burn like the Apollo's mission.

References

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